



Estimation Techniques

Predicting the Unpredictable

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🚀 **TL;DR**

We can't tell the future, but we can calculate it using data.

Extrapolation: Using history to predict the future.

Ratio Method: Using standard percentages (e.g., Test time vs. Dev time).

Three-Point Estimation: Using math to account for risk.

🤖 **The Core Challenge**

"How long will it take to test this?"

This is the hardest question a tester faces. If you say "2 days" and it takes "5 days," the release is delayed.

We use structured, data-driven techniques to defend our answers.

The Data-Driven Techniques

1. Extrapolation (Historical Data)

The Logic: "History Repeats Itself."

We look at previous projects that were similar to the current one.

- **Formula:** $(\text{Time taken for Old Project} / \text{Size of Old Project}) * \text{Size of New Project}$
- **Example:** Last month, testing **20** Test Cases took **30** hours.
- **Prediction:** The new feature has **100** Test Cases.
- - Calculation: $(30 / 20) * 100 = \mathbf{150 \text{ Hours.}}$

2. Estimation by Ratio

The Logic: "The Developer Rule."

We estimate testing effort based on development effort. Standard industry ratios often range from 30% to 50% of dev time.

- **Formula:** $\text{Dev Time} * \text{Ratio}$
- **Example:** The developers estimate the login feature will take **200 hours** to build.
- **Prediction:** Based on our team's standard ratio (40%), testing will take:
- - Calculation: $200 * 0.40 = \mathbf{80 \text{ Hours.}}$

3. Three-Point Estimation (The Risk Assessment)

The Logic: "Hope for the best, prepare for the worst."

We calculate an average based on three scenarios (Optimistic, Pessimistic, Most Likely).

- **Formula:** $(\text{Optimistic} + 4 * \text{Likely} + \text{Pessimistic}) / 6$
 - **Example:**
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 - Best Case: 2 hours.
 - Most Likely: 5 hours.
 - Worst Case: 14 hours.
 - Calculation: $(2 + 20 + 14) / 6 = \mathbf{6 \text{ Hours.}}$
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The Agile Technique

Planning Poker (Wideband Delphi)

The Logic: "Team Consensus."

Sometimes data isn't enough. The whole team gathers. Everyone secretly picks a number (Fibonacci card). We reveal them at once.

- If I say **2 days** and the Dev says **8 days**, we discuss the difference immediately.
 - This prevents one loud person from deciding for everyone.
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Real World Example: Migration Project

The Task: We are moving the database from System A to System B.

Using Extrapolation:

"Last year, migrating the 'User Table' (1000 rows) took 5 hours. The 'Product Table' has 5000 rows. Therefore, it should take **25 hours.**"

Using Ratio:

"The developer spent 10 hours writing the migration script. My testing ratio is usually 50% for backend tasks. I need **5 hours** to verify it."

Why This Matters

Estimations set expectations.

If I just guess, I look like an amateur.

If I say, *"Based on historical data from our last sprint, I calculate 150 hours,"* I look like a professional Test Manager.